

Sample Surveys

Class 4

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Previously
on
Sample Surveys...

Recap

- Estimation of population mean $\bar{Y}$ under SRSWOR
 - Estimator: Sample mean $\bar{y}$. $\bar{y}$ is unbiased for $\bar{Y}$.
 - Sampling variance of $\bar{y}$, $\text{Var}(\bar{y}) = (1 - f) \frac{S^2}{n}$.
 - $1 - f$: Reduction in variability between samples that happens due to different samples having some common units.
 - Estimator of sampling variance: $\widehat{\text{Var}}(\bar{y}) = (1 - f) \frac{s^2}{n}$.
- For large N , SRSWR: a quicker alternative to SRSWOR.
- Application of SRSWR: Bootstrap methods.
- Other estimands: Population total, population variance, population covariance (between two study variables).
- Estimating population proportions: Work with a binary Y , indicating the characteristic of interest.
- Measures of accuracy: sampling variance, standard errors, coefficient of variation or relative standard deviation.
- Another estimation method: SRSWR + use distinct units only.
 - Effective sample size: n_{eff}
 - The sample mean $\bar{y}_{\text{eff}}$ is unbiased for $\bar{Y}$.
 - Is it better than usual SRSWR with n units?

Confidence interval

- **Interval estimation** of the parameter of interest.
- $100(1 - \alpha)\%$ confidence interval: A **random interval** $[L, U]$ that contains the parameter of interest with probability at least $1 - \alpha$.

Canonical case: IID sampling from Normal distribution

- $X_1, X_2, \dots, X_n \stackrel{iid}{\sim} \mathcal{N}(\mu, \sigma^2)$, σ known (say). Estimator of μ : $\hat{\mu} = \bar{X}$.
 - $\bar{X} \sim \mathcal{N}(\mu, \frac{\sigma^2}{n})$ (sampling distribution). $SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}$.
 - $100(1 - \alpha)\%$ confidence interval (CI) for μ : $[\bar{X} - z_{\frac{\alpha}{2}} \times SE(\bar{X}), \bar{X} + z_{\frac{\alpha}{2}} \times SE(\bar{X})]$.
 - $z_{\frac{\alpha}{2}}$: $(1 - \frac{\alpha}{2})$ th quantile of $\mathcal{N}(0, 1)$ distribution
 - For $\alpha = 0.05$, $z_{\frac{\alpha}{2}} \approx 1.96$.
 - 95% CI for μ : $[\bar{X} - 1.96 \times \frac{\sigma}{\sqrt{n}}, \bar{X} + 1.96 \times \frac{\sigma}{\sqrt{n}}]$.
 - **Coverage probability** of the above random interval: 0.95.
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- 95% CI for $\bar{Y}$ in finite population sampling:
 - If we independently draw many random samples from the finite population, at least 95% of the resulting random intervals will contain the true value of $\bar{Y}$.

Confidence interval in finite population sampling

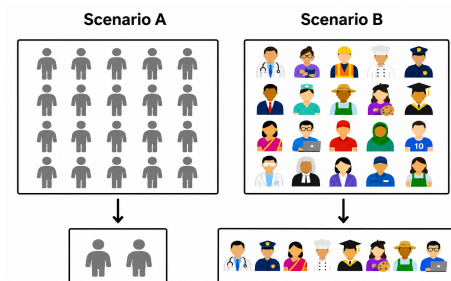
- In SRS, the sampling distribution of $\bar{y}$ depends on $Y_1, \dots, Y_N$ (**unknown**).
 - If we knew the exact sampling distribution, we would be able to compute the exact coverage probability of any random interval.
- **Cannot get an exact 95% CI for $\bar{Y}$ based on $\bar{y}$.**
- Need to rely on **asymptotic approximations**.
- **Rough idea:** Hájek (1960) showed that, under some **regularity conditions** on the finite population, and if N , n , and $(N - n)$ are **sufficiently large**,

$$\frac{\bar{y} - \bar{Y}}{\text{SE}(\bar{y})} \stackrel{\text{approx}}{\sim} \mathcal{N}(0, 1).$$

- That is, sampling distribution of $\bar{y}$ can be approximated by a Normal distribution.
- **Approximately**, $[\bar{y} - 1.96 \times \text{SE}(\bar{y}), \bar{y} + 1.96 \times \text{SE}(\bar{y})]$ contains the population mean with probability 0.95.
- Plug in the estimate of the standard error.
- Approx 95% CI for $\bar{Y}$: $[\bar{y} - 1.96 \times \widehat{\text{SE}}(\bar{y}), \bar{y} + 1.96 \times \widehat{\text{SE}}(\bar{y})]$.
- For small n , t_{n-1} -distribution gives a better approximation to the sampling distribution.

How to choose the sample size n ?

- One of the most important practical problems in sample surveys.
 - Too small n : Estimate may be completely **unreliable**.
 - Too large n : Very **costly** and **time consuming**. Often you have a limited budget.
 - “Bigger is always better” – not always true (**non-sampling errors**).
 - What n is **just right**?
- Assume no non-sampling errors for now.
- **No one size fits all formula.**



Determination of n : a recipe

- Ask yourself: What do I want to achieve with my sample?
 - How much precision do I need?
 - How much error will I tolerate?
- Don't think about: what percentage of the population should I include in my sample.
- Find an equation that relates n with your expectations of the sample.
- Solve for n . (not straightforward though)
- If the resulting n is much larger than you can afford, adjust your expectations.
- If you can't even come close to the precision you want with the resources you have, think whether you should even conduct the survey.

Margin of error

- Determines how much precision is needed and how much error is tolerable.
- In absolute terms: $P(|\bar{y} - \bar{Y}| \leq e) = 1 - \alpha$.
 - e : Absolute margin of error.
- In relative terms: $P\left(\left|\frac{\bar{y} - \bar{Y}}{\bar{Y}}\right| \leq r\right) = 1 - \alpha$.
 - r : Relative margin of error.
- You specify e (or r), and find an appropriate n .
 - Connect to the approximate $100(1 - \alpha)\%$ CI.
- What if we don't specify a margin of error?
 - E.g., we want to estimate the average monthly rent for a Paying Guest accommodation in Baranagar. Suppose we set $n = 15$.
 - Get CI: ₹6000 ± ₹1265, i.e., [₹4735, ₹7265].
 - What we should do instead: Set e , e.g., $e = ₹300$.
- Another example: Suppose we want to estimate the mean birth weight of infants born to mothers who smoke cigarettes during pregnancy.
 - Margin of error should be pretty small, i.e., 0.9 kg.

Determining n for SRS

- WLOG, focus on absolute margin of error e .
- For SRSWR, solving for n , we get: $n_{WR} = \frac{z_{\frac{\alpha}{2}}^2 \sigma^2}{e^2}$.
- For SRSWOR, solving for n , we get: $n_{WOR} = \frac{z_{\frac{\alpha}{2}}^2 S^2}{e^2 + \frac{z_{\frac{\alpha}{2}}^2 S^2}{N}}$.
- $n_{WOR} \approx \frac{n_{WR}}{1 + \frac{n_{WR}}{N}}$.
- But n_{WR} and n_{WOR} depend on **unknown population quantities**.

Special case: Estimation of population proportion

- Parameter of interest: P .
- $n_{WR} = \frac{z_{\frac{\alpha}{2}}^2 P(1-P)}{e^2}$, $n_{WOR} \approx \frac{z_{\frac{\alpha}{2}}^2 P(1-P)}{e^2 + \frac{z_{\frac{\alpha}{2}}^2 P(1-P)}{N}}$.
- Note that $P(1-P) \leq \frac{1}{4}$. Put it back in the formula.
- Get n corresponding to the **worst-case** scenario.